



Kushagra Agarwal

BTech

Chemical Engineering

Indian Institute Of Technology, Tirupati

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[LinkedIn](#)

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. (CH)	Indian Institute of Technology, Tirupati	8.49	2022-2026
Senior Secondary	Aligarh Muslim University Board	96.0%	2022
Secondary	Aligarh Muslim University Board	97.5%	2020

EXPERIENCE

- Bureau of Indian Standards** June 2025 - July 2025
Summer Intern Vijayawada, India
 - Compared quality standards and compliance between large-scale and MSME manufacturing units.
 - Helped prepare technical proposals for new product standards by researching process automation, testing methods, and industry applications.
 - Conducted retail compliance audits and created reports to improve enforcement and ensure consumer safety.
 - Gained in-depth knowledge of Indian Standards and regulatory frameworks, enhancing understanding of quality and compliance requirements.

PROJECTS

- Hydrodealkylation of Toluene to Produce Benzene** Jan 2024 - May 2025
Worked under Dr. Shamik Misra IIT Tirupati
 - Worked on the design and simulation of reactor, flash distillation, and heat transfer units for the hydrodealkylation process.
 - Focused on optimizing process efficiency and energy consumption in the production of benzene from toluene.
 - Applied knowledge of chemical reaction engineering, thermodynamics, and heat transfer to ensure smooth operation and product quality.
 - Performed ASPEN Plus simulation of chemical looping combustion of syngas in fluidized beds and air reactors.
 - Tools & technologies used:** MS Excel, MATLAB, MS PowerPoint, Python, ASPEN Plus
- Development and Electrochemical Analysis of NASICON-type Cathode Materials** Aug 2025 - Present
Worked under Dr. Ranjith Thangavel IIT Tirupati
 - Synthesized and optimized NASICON-type cathode materials — NVP ($\text{Na}_3\text{V}_2(\text{PO}_4)_3$), Ti-doped NVTP, and Ti-Cr co-doped NVTCP for sodium-ion batteries.
 - Conducted material characterization using XRD, FTIR, and TGA to study structural, functional and thermal properties.
 - Evaluated electrochemical performance through CV, GCD and EIS observing improved conductivity and cycling stability in doped samples.
 - Tools & technologies used:** Tools techniques used: XRD, FTIR, TGA, CV, GCD, EIS, Origin and solid-state synthesis.

TECHNICAL SKILLS

- Programming & OS:** Python, Linux, Windows
- Tools:** Scikit learn, Pandas, Google Colab, MATLAB, Aspen Plus, AutoCAD, Simulink, Origin

RELEVANT COURSES

- Courses:** Fluid and Particle Mechanics, Chemical Engineering Thermodynamics, Process Heat Transfer, Reaction Engineering, Mass Transfer, Computational Techniques, Process Control and Instrumentation, Engineering Economics and Sustainability, Electrochemical Energy Storage, Battery Manufacturing Technology

LABORATORY SKILLS

- Labs:** Fluid and Particle Mechanics Laboratory, Reaction Engineering Lab, Process Design and Simulation Laboratory, Process Heat Transfer Laboratory, Mass Transfer Laboratory, Computational Techniques Lab(ASPEN Plus, MATLAB, SIMULINK)

POSITIONS OF RESPONSIBILITY

- Club Head** , Chemical Engineering Club, IIT Tirupati Aug 2024 - May 2025
- Finance Committee** , Tirutsava, IIT Tirupati Aug 2024 - May 2025
- Student Placements Representative** , CDC, IIT Tirupati Aug 2025 - Present

CERTIFICATIONS

- [Supervised Machine Learning: Regression and Classification \(Stanford\)](#)
- [Python for Data Science and AI \(IBM\)](#)
- [Inherently Safer Design](#)
- [Hydrogen Energy](#)